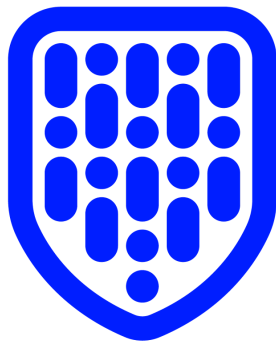


Capstone Project

Customer Churn Prediction for Telecommunications Company



**Institute_{of}
Data**

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Date : 14 April 2025

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Problem Statement

Customer churn refers to the termination of a customer's relationship with a service provider and represents a critical business metric in the telecommunications industry due to its direct impact on recurring revenue and long-term profitability. Unlike one-time transactional businesses, telecom services significantly rely on continuous customer subscriptions to generate stable and predictable revenue. Therefore, every customer who churns generates immediate revenue loss and forfeits potential future earnings.

Customer churn is thus more than a customer service issue; it is a fundamental business challenge. Acquiring new customers requires substantial investments in marketing, onboarding, and operational resources, making retention significantly more cost-effective. Forbes cites research indicating that boosting customer retention by just 5% can increase profit by 25% to 95%. Effective retention strategies increase customer lifetime value (CLV), strengthen brand loyalty, improve customer satisfaction, and enable telecom providers to maintain and expand their market share in a highly competitive environment.

Industry / Domain

Customer Churn Across Different Industries

Telecommunications:

Services rely heavily on ongoing customer subscriptions for predictable income. Customer churn leads directly to revenue loss and forfeited future earnings.

Software as a Service (SaaS):

Customer churn occurs when subscriptions are canceled, negatively impacting revenue growth and profitability. Churn in SaaS often results from perceived lack of value, usability issues, or competitive alternatives. Thus, SaaS companies prioritize customer success, usability improvements, and continuous value delivery.

Banking:

In banking, churn involves customers closing accounts or switching to other financial institutions, frequently due to dissatisfaction with service quality, fees, or more attractive offers from competitors. Given high acquisition costs and the importance of ongoing revenue streams such as loans, credit cards, and investments, customer retention in banking is crucial.

Overall, customer churn is a significant challenge across these sectors, making proactive retention strategies essential for sustained profitability.

Impact of SIM-only Plans on Churn Risk

In Singapore's telecommunications landscape, a significant shift has occurred since 2015. Previously, telco providers primarily relied on bundled mobile data plans with multi-year contracts offering subsidized phones, effectively locking consumers into their services. However, today's market presents a vastly different scenario where consumers can easily migrate between telco providers to secure deals that best align with their lifestyle needs and budget constraints.

This market transformation has substantially empowered consumers, enabling more informed decision-making while simultaneously pressuring telcos to continually innovate and enhance their service offerings to maintain their customer base in an increasingly competitive environment.

The growing popularity of SIM-only plans, particularly those offered by Mobile Virtual Network Operators (MVNOs), has introduced additional churn risk factors by providing customers unprecedented flexibility with minimal contractual obligations. This development has significantly increased customer mobility, facilitating easier provider transitions that ultimately contribute to elevated churn rates and revenue losses for established providers.

Further complicating retention efforts is the telecommunications industry's persistent reliance on reactive customer service models. By responding only after customers have already decided to leave, providers miss critical opportunities to address early behavioral indicators that signal potential churn. This reactive approach substantially diminishes the effectiveness of retention strategies by failing to identify and address customer dissatisfaction at stages where intervention could still prove successful.

Necessity of Proactive Retention Strategies

To remain competitive and nurture lasting relationships, telecom companies must adopt proactive, data-driven retention strategies. Churn prediction models utilizing data science techniques have shown effectiveness in identifying customers at risk of churn by analyzing historical and behavioral data. This enables providers to proactively address churn drivers, enhancing customer retention, safeguarding revenue, and increasing long-term customer value.

Transition from Reactive to Predictive Customer Management

Transitioning from reactive customer management to a predictive, proactive approach through churn prediction models allows early detection of potential churn. Timely interventions based on these predictions significantly improve the customer experience and generate positive business outcomes.

Academic and Industry Approaches to Churn Prediction

Churn prediction has been extensively studied using various machine learning and statistical methods, including logistic regression, decision trees, random forests, XGBoost, and neural networks. The effectiveness of these models varies based on dataset characteristics, feature engineering, and specific business contexts. Optimal models typically depend on organizational objectives and churn indicators like tenure, billing, service usage, complaint history, and contract types.

Predictive analytics empowers telecom providers to anticipate customer churn resulting from insufficient communication, unresolved complaints, or competitive offers. Integrating predictive models into CRM systems has proven successful in enabling timely interventions and targeted retention campaigns, significantly enhancing loyalty and reducing churn.

However, model interpretability, data quality, and actionable insights remain challenging. Complex models, such as ensemble and deep learning algorithms, often lack transparency. Explainer tools like SHAP and LIME address this by clarifying individual feature contributions, thus improving stakeholder confidence and enabling actionable retention strategies.

Stakeholders

This project benefits stakeholders including:

- **Customer Retention and Success Teams:** Seeking proactive engagement solutions for at-risk customers.
- **Marketing Managers:** Aiming to enhance campaign effectiveness and reduce churn.
- **CRM Administrators:** Integrating predictive analytics into customer management systems.
- **Data Analysts and Scientists:** Deriving actionable insights from data.
- **Senior Management and Executives:** Making strategic decisions focused on revenue protection and profitability.
- **Product Managers:** Refining customer experience and service offerings.

Collectively, these stakeholders use predictive insights to enhance customer satisfaction, reduce churn rates, and sustain profitability.

The insights derived from this churn prediction model will enable stakeholders to proactively identify and engage customers at high churn risk, tailor retention campaigns effectively, and improve customer satisfaction through targeted interventions. Marketing teams can design personalized promotions, CRM administrators can automate churn-risk notifications, and senior management can make informed strategic decisions to optimize retention efforts. Ultimately, these actions will enhance customer loyalty, preserve revenue streams, and significantly contribute to sustained profitability and competitive advantage.

Business Question

As a Business Analyst at a telecom company, this project focuses on analyzing the underlying drivers of customer churn and building a predictive model to identify customers at risk of churning in the upcoming billing cycle. The analysis aims to enable proactive, data-driven retention strategies that reduce revenue loss and improve customer lifetime value.

Business Value:

Reducing churn by even 5% could significantly increase annual revenue. Assuming an average monthly subscription of \$50 and a customer base of 100,000, a 5% reduction in churn (5000 customers) would result in approximately \$3 million (SGD) in annual retained revenue.

Required Accuracy and Implications:

The prediction model should achieve at least 85% accuracy. High accuracy is crucial because:

- **False Positives:** Unnecessary retention measures may lead to wasted resources and increased operational costs.
- **False Negatives:** Missed churn cases result in lost revenue and reduced customer lifetime value.

Balancing these outcomes is critical to maximize overall business performance and profitability.

Data Question

This project aims to predict customer churn by analyzing customer demographics and behavioral data using machine learning techniques. By identifying key factors that influence a customer's decision to leave, the analysis evaluates the effectiveness and accuracy of various predictive models. The insights gained from this project can help businesses proactively address customer churn, enabling them to implement targeted strategies to retain valuable customers and enhance overall customer satisfaction.

Data

The original dataset is from IBM and sourced from Kaggle. The dataset contains 7,043 rows and 33 columns.

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7043 entries, 0 to 7042
Data columns (total 33 columns):
#   Column              Non-Null Count  Dtype  
---  -
0   CustomerID          7043 non-null   object  
1   Count                7043 non-null   int64   
2   Country              7043 non-null   object  
3   State                7043 non-null   object  
4   City                 7043 non-null   object  
5   Zip Code             7043 non-null   int64   
6   Lat Long             7043 non-null   object  
7   Latitude             7043 non-null   float64  
8   Longitude            7043 non-null   float64  
9   Gender               7043 non-null   object  
10  Senior Citizen       7043 non-null   object  
11  Partner              7043 non-null   object  
12  Dependents           7043 non-null   object  
13  Tenure Months        7043 non-null   int64   
14  Phone Service        7043 non-null   object  
15  Multiple Lines        7043 non-null   object  
16  Internet Service     7043 non-null   object  
17  Online Security      7043 non-null   object  
18  Online Backup         7043 non-null   object  
19  Device Protection    7043 non-null   object  
20  Tech Support         7043 non-null   object  
21  Streaming TV         7043 non-null   object  
22  Streaming Movies     7043 non-null   object  
23  Contract             7043 non-null   object  
24  Paperless Billing     7043 non-null   object  
25  Payment Method       7043 non-null   object  
26  Monthly Charges      7043 non-null   float64  
27  Total Charges        7043 non-null   object  
28  Churn Label          7043 non-null   object  
29  Churn Value          7043 non-null   int64   
30  Churn Score          7043 non-null   int64   
31  CLTV                 7043 non-null   int64   
32  Churn Reason         1869 non-null   object  
dtypes: float64(3), int64(6), object(24)
memory usage: 1.8+ MB

```

Fig 1 : Overview of dataset

Data Science Process

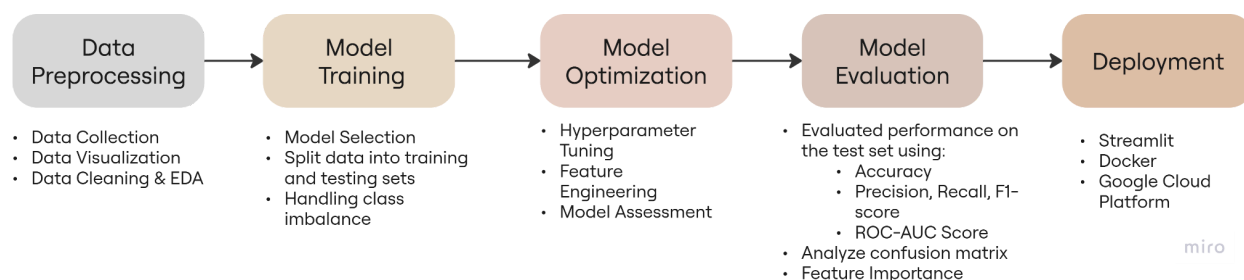


Figure 2 : Flowchart of developing customer churn prediction machine learning model

The flowchart above illustrates an overview of the data science pipeline adopted to develop a machine learning model.

Data Preprocessing

This is the foundational phase where raw data is transformed into a clean and structured format suitable for modeling.

- **Data Collection:** Gathered relevant data from telecom records, including customer demographics, service usage patterns, contract details, and churn labels.

Upon reviewing the dataset gathered from Figure 1, the features can be categorised into these 5 categories as follows:

Feature Category

Customer Info: CustomerID, Gender, Senior Citizen, Partner, Dependents

Service Details: Phone Service, Internet Service, Tech Support, Streaming TV

Account Details: Tenure Months, Contract, Payment Method, Paperless Billing

Billing: Monthly Charges, Total Charges, CLTV

Churn Indicators: Churn Label, Churn Value, Churn Reason, Churn Score

Key columns include:

- Churn Label: Indicates whether a customer has churned ("Yes" or "No").
- Churn Value: Numeric (1 for churned, 0 for not).
- Churn Reason: Text reason for churn (only available for 1,869 customers).

Data Visualization

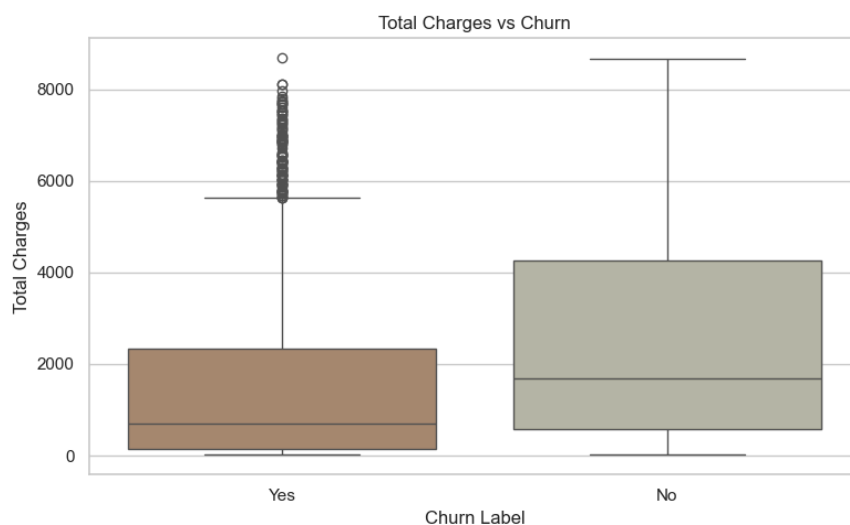


Figure 3 : Total Charges vs Churn

The Total Charges vs Churn analysis highlights a key trend: customers who did not churn generally have higher total charges, indicating longer tenure and stronger engagement. In contrast, those who churned often have lower total charges, suggesting they left early in their customer journey. This pattern shows that churn is more common among newer customers, while long-term users are more likely to stay. It underscores the importance of delivering a strong onboarding experience and early support to improve retention. Focusing on the initial customer lifecycle can significantly reduce churn and encourage long-term loyalty.

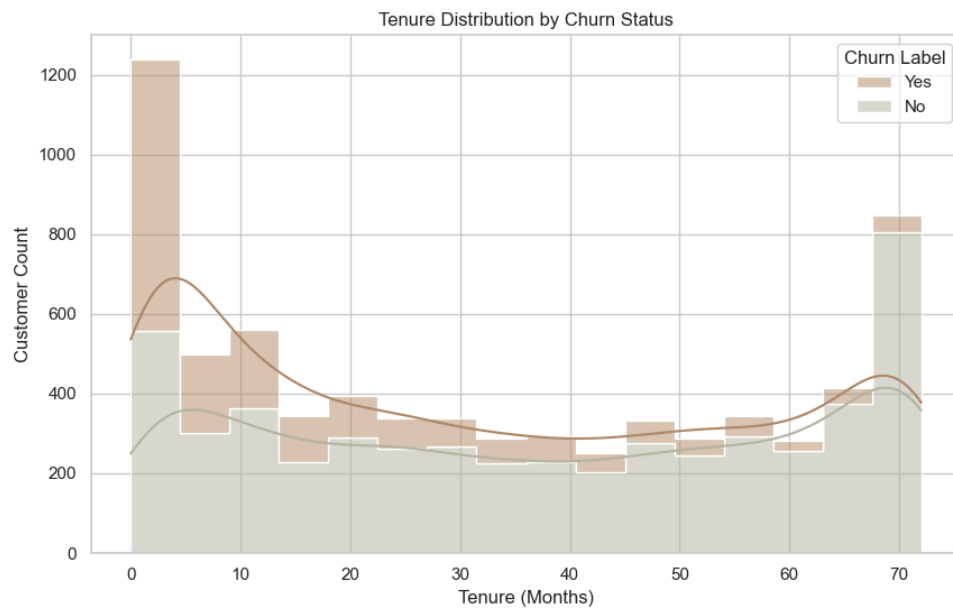


Figure 4 : Tenure Distribution by Churn Status

The Tenure Distribution by Churn Status shows that churn is most common among customers with shorter tenures. Many customers leave within the first few months, while those who stay longer—especially beyond 40 months—are far less likely to churn. This indicates that customer loyalty increases with time, and the risk of churn decreases as tenure grows. The insight emphasizes the importance of focusing on early-stage retention, ensuring a smooth onboarding experience and strong initial support to reduce early drop-off and build lasting customer relationships.

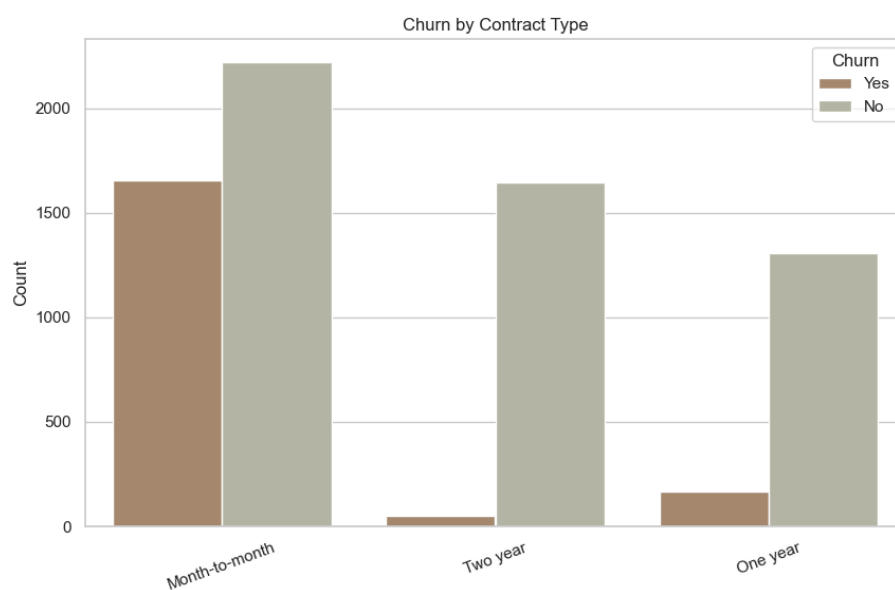


Figure 5 : Churn by Contract Type

The Churn by Contract Type analysis reveals that customers on month-to-month contracts have the highest churn rate, while those on one-year or two-year contracts are significantly more likely to stay. This suggests that longer-term contracts promote greater customer retention, likely due to stronger commitment, incentives, or reduced flexibility to switch providers. Month-to-month customers may feel less tied to the service and are more prone to leave if dissatisfied. This pattern highlights the value of encouraging customers to opt for longer contracts through loyalty benefits or personalized offers, which can help reduce churn and improve long-term retention.

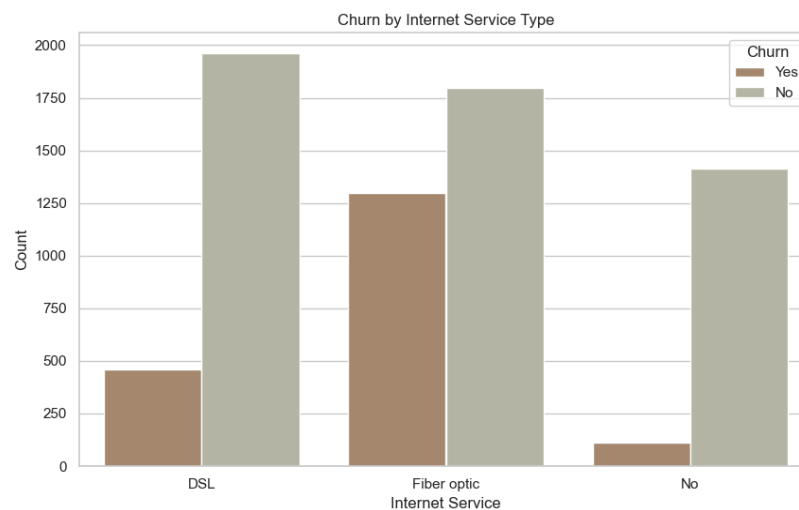


Figure 6 : Churn by Internet Service Type

The Churn by Internet Service Type analysis shows that customers with Fiber optic internet experience the highest churn rate compared to those with DSL or no internet service. This could be due to higher expectations around speed and reliability, or more competitive alternatives in the market. Customers without internet service show the lowest churn, possibly because their engagement with the provider is minimal or limited to other services. These findings suggest that churn mitigation efforts should focus on fiber optic users, ensuring service quality, responsive support, and value-added features to retain this

higher-risk customer group.

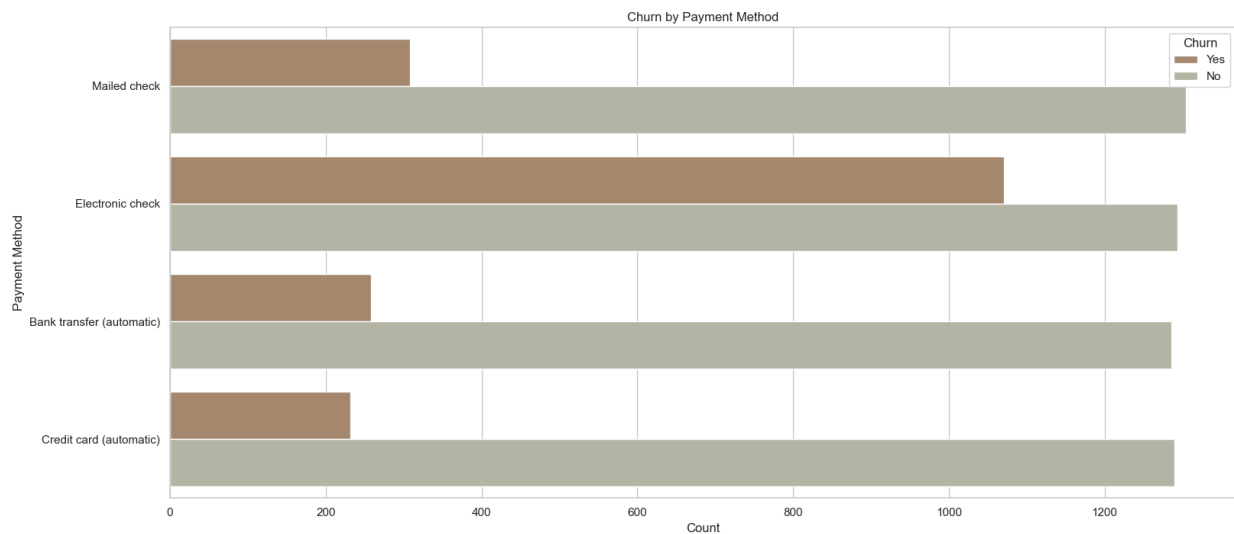


Figure 7 : Churn by Payment Method

The Churn by Payment Method analysis shows that customers using Electronic check have the highest churn rate, while those using automatic payment methods like bank transfers or credit cards are less likely to churn. This suggests that customers who choose auto-pay options may be more committed or experience fewer payment-related issues. The convenience and consistency of automatic payments can contribute to higher retention. Encouraging customers to switch to more stable payment methods could help reduce churn and improve overall customer loyalty by minimizing barriers to continued service.

Data Cleaning & Exploratory Data Analysis

Before the churn prediction model was built, several data cleaning and EDA steps were performed to ensure data quality and extract meaningful insights:

1. Handling Missing Values:
 - The Total Charges column, although numeric in nature, contained blank or string values. These were converted from object to numeric using `pd.to_numeric(errors='coerce')`, and rows with missing values were removed for modeling.
2. Data Type Conversion:
 - Categorical features such as Senior Citizen, Partner, and Dependents were verified and converted to consistent types (e.g., strings or integers) to ensure

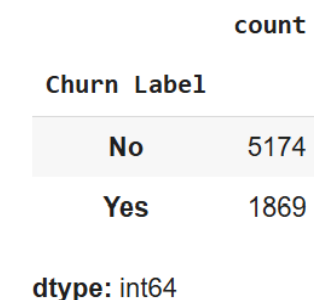
proper encoding.

3. Target Variable Identification:

- The churn indicator was clearly defined using the Churn Label and Churn Value columns. For modeling, Churn Label(0 = No, 1 = Yes) was used as the binary target variable.

4. Exploratory Visualizations:

- Referring to figure 8, churn distribution was visualized to reveal **class imbalance** (fewer churned customers), which was flagged as a key consideration for model evaluation
- With a higher number of non-churned customers, the dataset is imbalanced. This insight informed decisions about performance metrics and potential resampling strategies during modeling.



count	
Churn Label	
No	5174
Yes	1869

. dtype: int64

Figure 8 : Insights on churn distribution

- Relationships between features and churn were explored using bar plots, histograms, and boxplots (e.g., Total Charges vs. Churn, Tenure vs. Churn).
- A correlation heatmap was created to assess relationships between numerical features, revealing strong positive correlations between **tenure and total charges**, and moderate negative correlation between **tenure and churn**.

5. Feature Reduction & Selection :

Features like CustomerID, Zip Code, and precise Latitude/Longitude were deemed less useful for prediction and hence was dropped. .

- Strong predictors like Contract, Tenure, Monthly Charges, and Payment Method were identified for use in model training.

Correlation Map

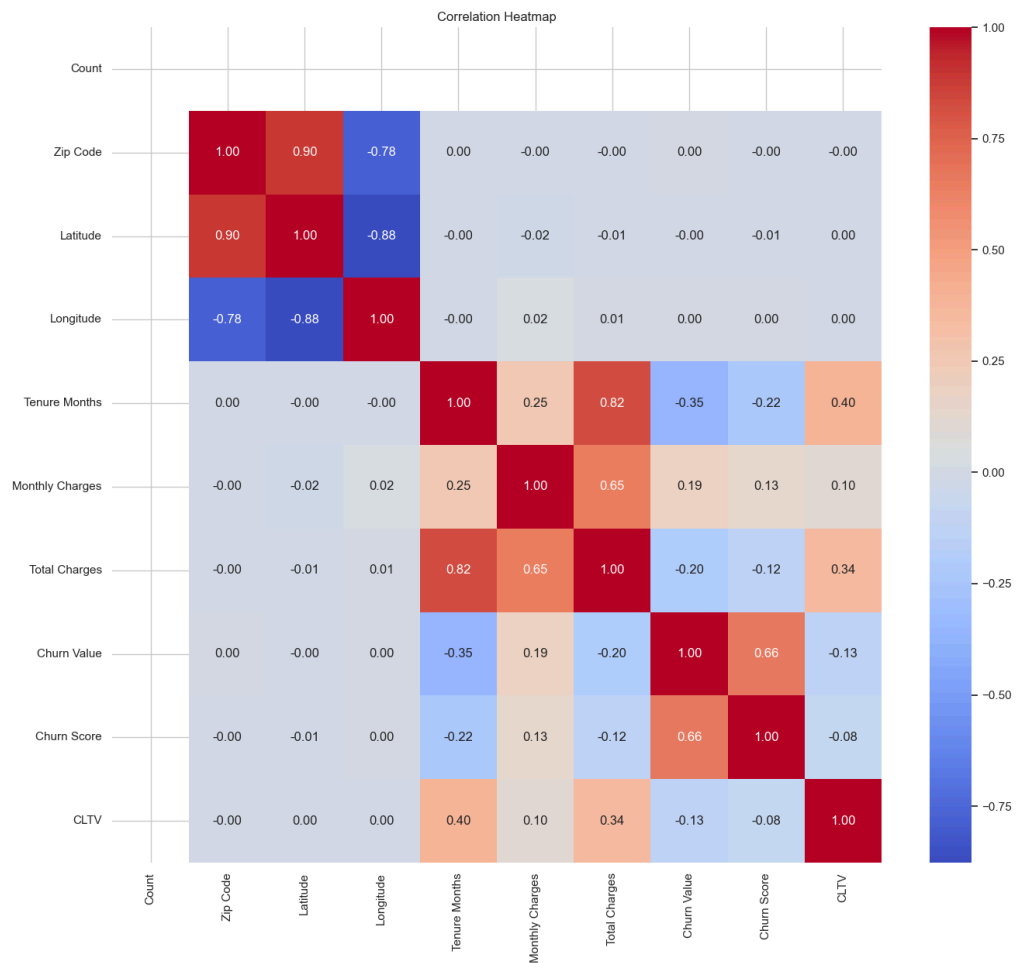


Figure 8 : Correlation Map

Correlation of Numerical Features with Churn Indicators

Tenure Months & Total Charges (0.82)

- Strong positive correlation: the longer a customer stays, the more they are likely to spend. This makes sense, as tenure accumulates charges over time.

Churn Value & Contract (-0.40 to -0.46)

- Moderate negative correlation with tenure and total charges: churned customers tend to have **shorter tenures and lower total charges**.

Monthly Charges & Total Charges (0.65)

- Moderate correlation: customers paying more monthly tend to accumulate higher total charges, depending on how long they've stayed.

Churn Score & Churn Value (0.52)

- Positively correlated: higher churn scores align with customers who actually churned, suggesting the score is a good indicator.

CLTV (Customer Lifetime Value) & Total Charges (0.41)

- Moderate positive correlation: high-value customers tend to generate more revenue, though it's not a perfect relationship.

Model Training

To prepare the dataset for machine learning, the following preprocessing steps were applied:

1. Dropping Irrelevant Features:

Columns such as CustomerID, Lat Long, Zip Code, Latitude, and Longitude were dropped, as they do not provide predictive value and may introduce noise.

2. Handling Categorical Variables:

Categorical features (e.g., Contract, Payment Method, Internet Service, etc.) were encoded using **one-hot encoding** to convert them into numerical format, allowing them to be used by most machine learning models.

3. Encoding Binary Variables:

Binary variables such as Partner, Dependents, Phone Service, and Paperless Billing were encoded as 0 (No) and 1 (Yes) for simplicity and consistency.

4. Feature Scaling:

Numerical features like Monthly Charges, Total Charges, and Tenure Months were standardized or normalized to ensure they are on a similar scale, especially important for models sensitive to feature magnitude (e.g., logistic regression, SVM).

5. Handling Class Imbalance:

Since the dataset had more non-churned than churned customers, techniques such as SMOTE (Synthetic Minority Oversampling Technique) or upscaling were considered to ensure the model learns from both classes effectively.

6. Train-Test Split:

The data was split into **training and testing sets** (e.g., 80/20 or 70/30) to evaluate the model's ability to generalize to unseen data.

7. Feature Selection:

Features with low correlation to churn or high multicollinearity were optionally removed or reviewed using **feature importance scores** from tree-based models or recursive feature elimination.

Model Evaluation

	Model	Accuracy	ROC AUC	Precision
3	Support Vector Machine	0.797729	0.713915	0.691030
2	Random Forest	0.794180	0.709174	0.683333
0	Logistic Regression	0.800568	0.730986	0.676558
5	XGBoost	0.786373	0.711267	0.649547
4	K-Nearest Neighbors	0.759404	0.692437	0.582656
1	Decision Tree	0.735983	0.679856	0.533981

Figure 9 : Evaluation of Machine Learning model results

- Logistic Regression achieved the highest accuracy (0.8005) and ROC AUC (0.7309), making it a strong overall performer, even compared to more complex models.
- Support Vector Machine delivered the highest precision (0.6910), which means it's slightly better at minimizing false positives (predicting churners correctly).
- Random Forest also performed well across all metrics and is a good balance of interpretability and power.
- KNN and Decision Tree underperformed, particularly in precision, meaning they're less reliable for identifying true churners.
- XGBoost was consistent but didn't outperform Random Forest or SVM in this case.

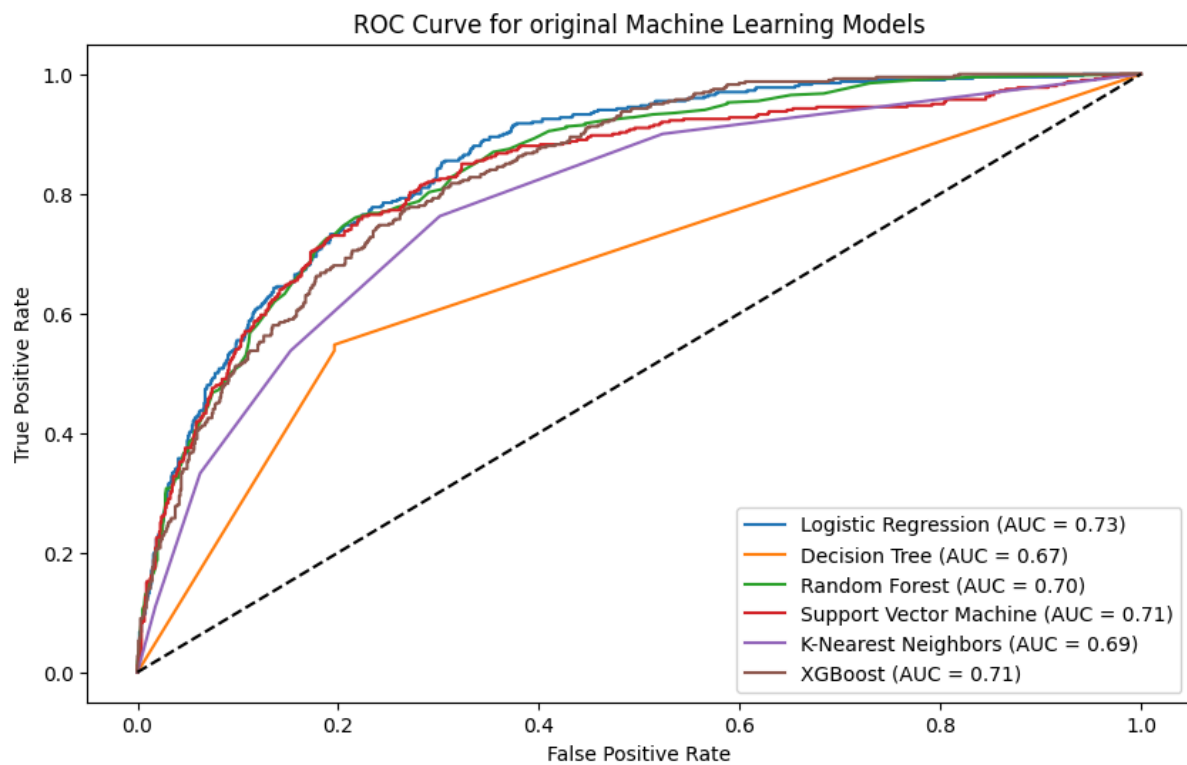


Fig 10. Results of using different machine learning models suitable for classification

Model Evaluation and Comparison

This analysis aimed to evaluate and compare the performance of two classification models typically used in predicting customer churn - Random Forest and Logistic Regression. The goal was to determine which model performs better in predicting the target variable, both in its original form and after applying techniques to address class imbalance and enhance model performance.

By comparing these models, we aimed to assess:

- The impact of model complexity on predictive performance.
- How each model responds to class imbalance handling (SMOTE, upsampling).
- If a simpler model (Logistic Regression) could offer competitive performance with proper tuning.

Interpretation of Results

- For **base models**, Logistic Regression slightly outperformed Random Forest (AUC 0.73 vs. 0.71).
- **Hyperparameter tuning** significantly improved both models:
 - Logistic Regression: AUC increased to 0.85.
 - Random Forest: AUC increased dramatically to 0.92 when combined with SMOTE or upsampling.
- **Class imbalance handling** (SMOTE/Upsampling) had a greater impact on Random Forest compared to Logistic Regression.
 - Logistic Regression saw no additional performance gain beyond tuning.
 - Random Forest's performance surged when class balancing was added.

Final Model Comparison

Given the results, the two models selected for final comparison were:

1. **SMOTE + Hypertuned Random Forest (AUC = 0.92)**
2. **Hypertuned Logistic Regression (AUC = 0.85)**

The decision to focus on these two was driven by:

- Their superior performance within their respective families.
- A meaningful contrast in complexity and interpretability.
- The desire to weigh model performance against the benefits of interpretability and computational cost.

Model Evaluation - Conclusion

- **Random Forest** with SMOTE and hyperparameter tuning was the best-performing model, achieving an AUC of 0.92.
- **Logistic Regression**, while simpler and more interpretable, peaked at an AUC of 0.85.

- Depending on the use case, either model could be justified:
 - Choose **Random Forest** for performance-critical applications.
 - Choose **Logistic Regression** when explainability and simplicity are paramount.

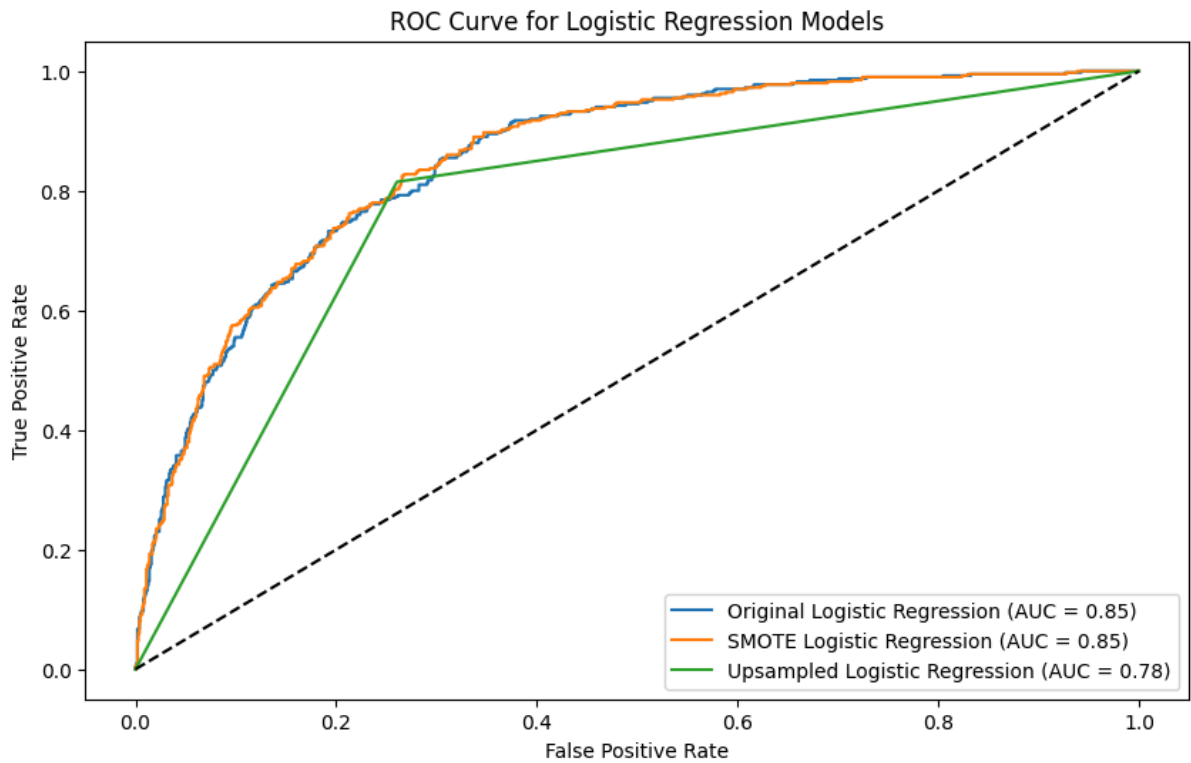


Figure 11: Results for Logistic Regression model after upscaling and hypertuning.

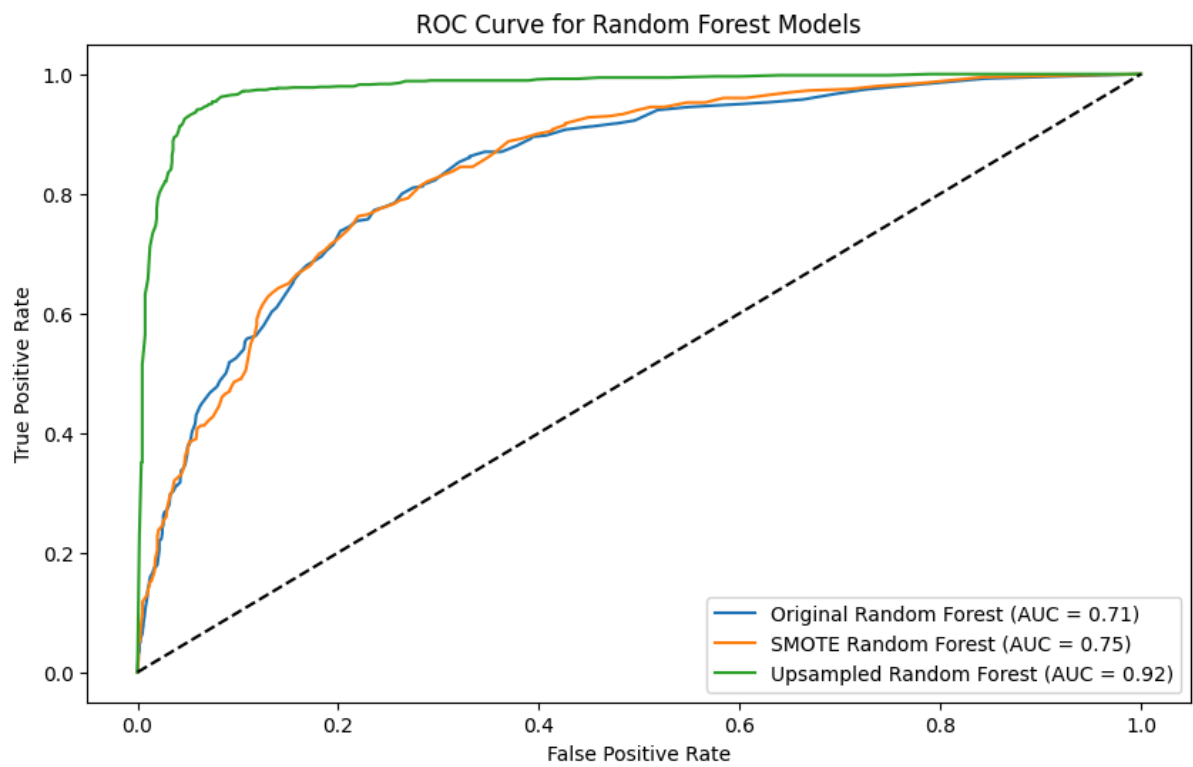
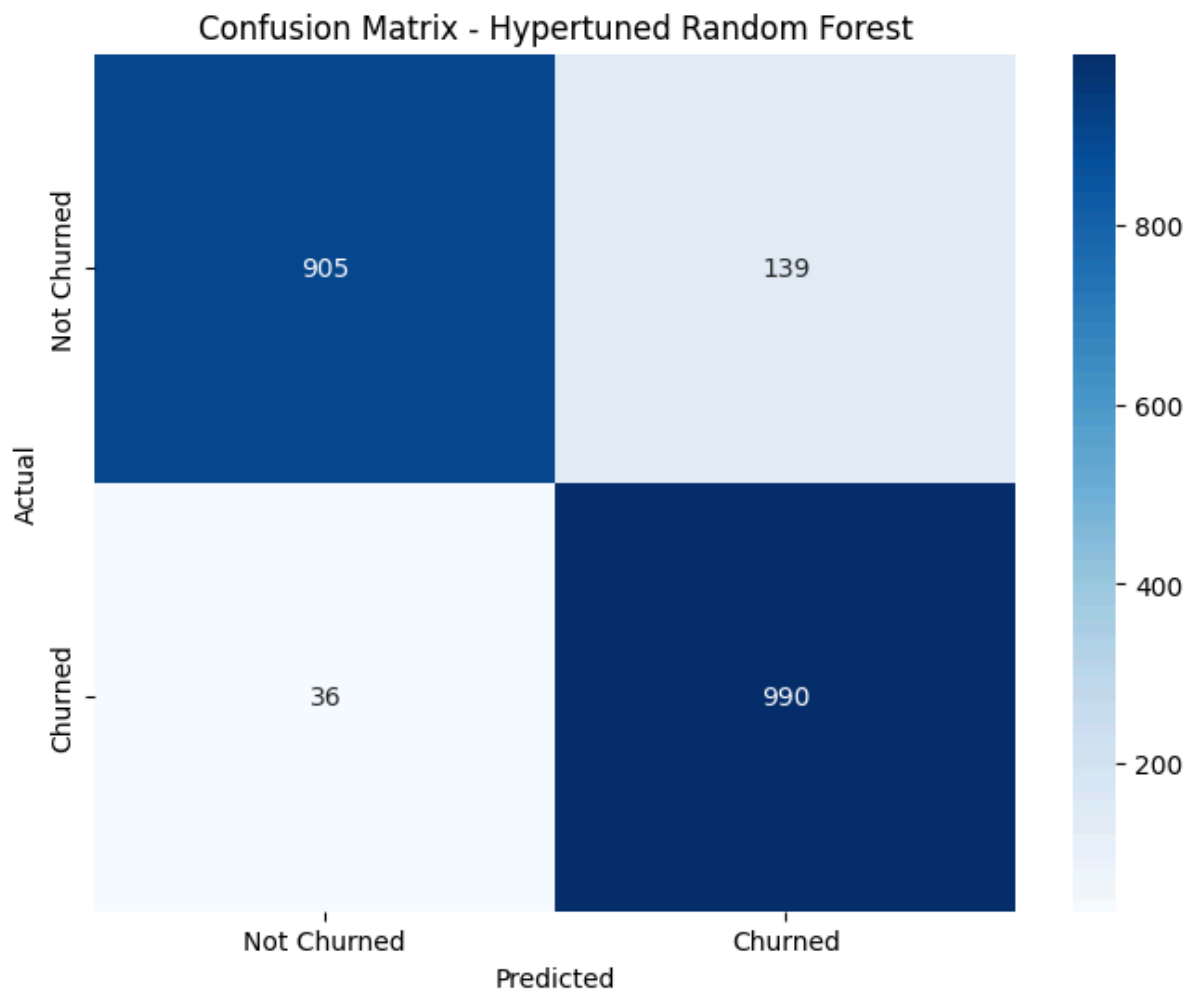


Figure 11 : Results for Random Forest + Upscaling + Hypertuning

Confusion Matrix



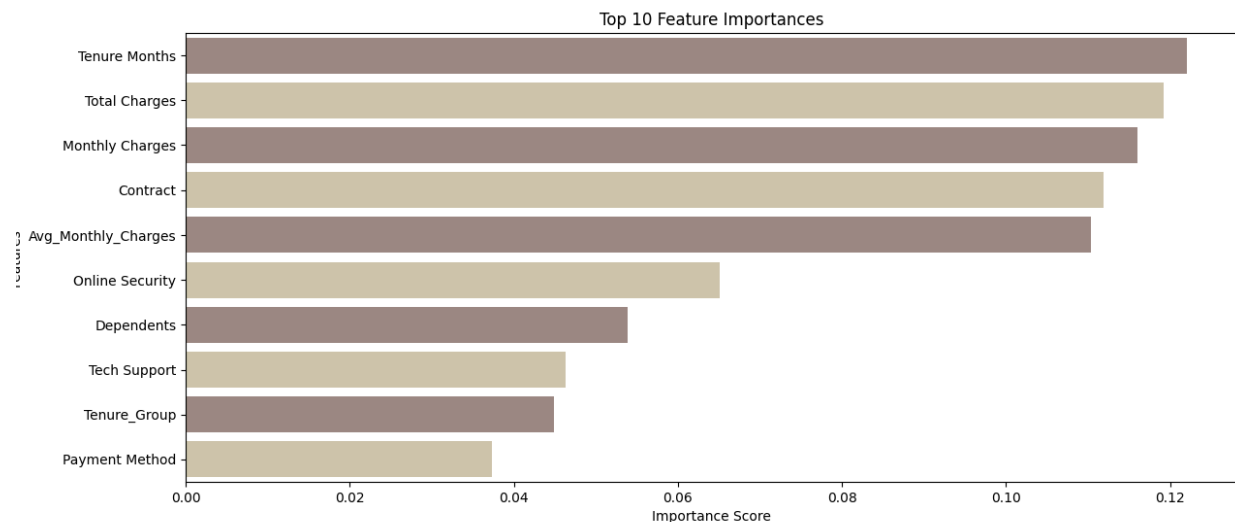
The confusion matrix shows the performance of the final Random Forest model in classifying churn and non-churn customers:

- **True Negatives (No Churn predicted correctly): 905**
- **False Positives (No Churn misclassified as Churn): 139**
- **False Negatives (Churn misclassified as No Churn): 36**
- **True Positives (Churn predicted correctly): 990**

This means the model has strong recall and precision, particularly for churn detection, which is the business-critical outcome.

There were only 43 churners were missed out of 1026 — a very low false negative rate, which is crucial when trying to retain customers.

Feature Importance



The feature importance analysis from the optimized Random Forest model revealed that tenure-related and financial variables play the most significant role in predicting churn. Tenure Months was the top feature, indicating that customers with shorter relationships are more likely to leave. This was closely followed by Total Charges and Monthly Charges, showing that both lifetime value and current billing influence churn behavior. The type of contract also had a major impact, with month-to-month customers being more likely to churn than those on longer-term agreements. Additional factors such as lack of online security or tech support, absence of dependents, and certain payment methods further contributed to churn risk. These insights can help businesses identify high-risk customers and tailor retention strategies accordingly.

Outcomes

The data science project resulted in the successful development of a high-performing customer churn prediction model. Using an optimized Random Forest classifier with upsampling and hyperparameter tuning, the model achieved an impressive ROC AUC of 0.92, demonstrating excellent predictive capability. The analysis identified key churn indicators such as short tenure (less than 12 months), month-to-month contracts, high monthly charges, and Electronic Check as a preferred payment method. These findings provide a strong foundation for targeted retention strategies. The final model was deployed via a Streamlit application on Google Cloud Platform (GCP), making it easily accessible for real-time use by business stakeholders.

Link : <https://churn-predictor-939566192323.asia-southeast1.run.app/>

Implementation

For production deployment, the model was successfully implemented as an interactive Streamlit web application hosted on GCP. This setup ensures broad accessibility and user-friendly interaction. Future integration with CRM systems is recommended to automate churn alerts and enable proactive retention outreach. Ongoing monitoring and periodic retraining will be essential to maintain performance and adapt to evolving customer behaviors. To further scale the solution, deploying the model as a REST API is a viable next step.

Data Answer

The machine learning model effectively addressed the data challenge by accurately identifying churn-prone customers based on historical patterns. With a strong ROC AUC score and interpretable features, the model offers high confidence in its predictions and supports data-driven decision-making for churn prevention.

Business Answer

From a business perspective, the model has delivered valuable insights into customer churn behavior and provides a practical tool for reducing attrition. Its deployment enables real-time prediction and actionable segmentation, allowing business teams to prioritize outreach efforts. Confidence in the model's business applicability is high, as the results align well with known patterns and provide clear direction for retention strategy.

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